

Algorithms Problem Solving

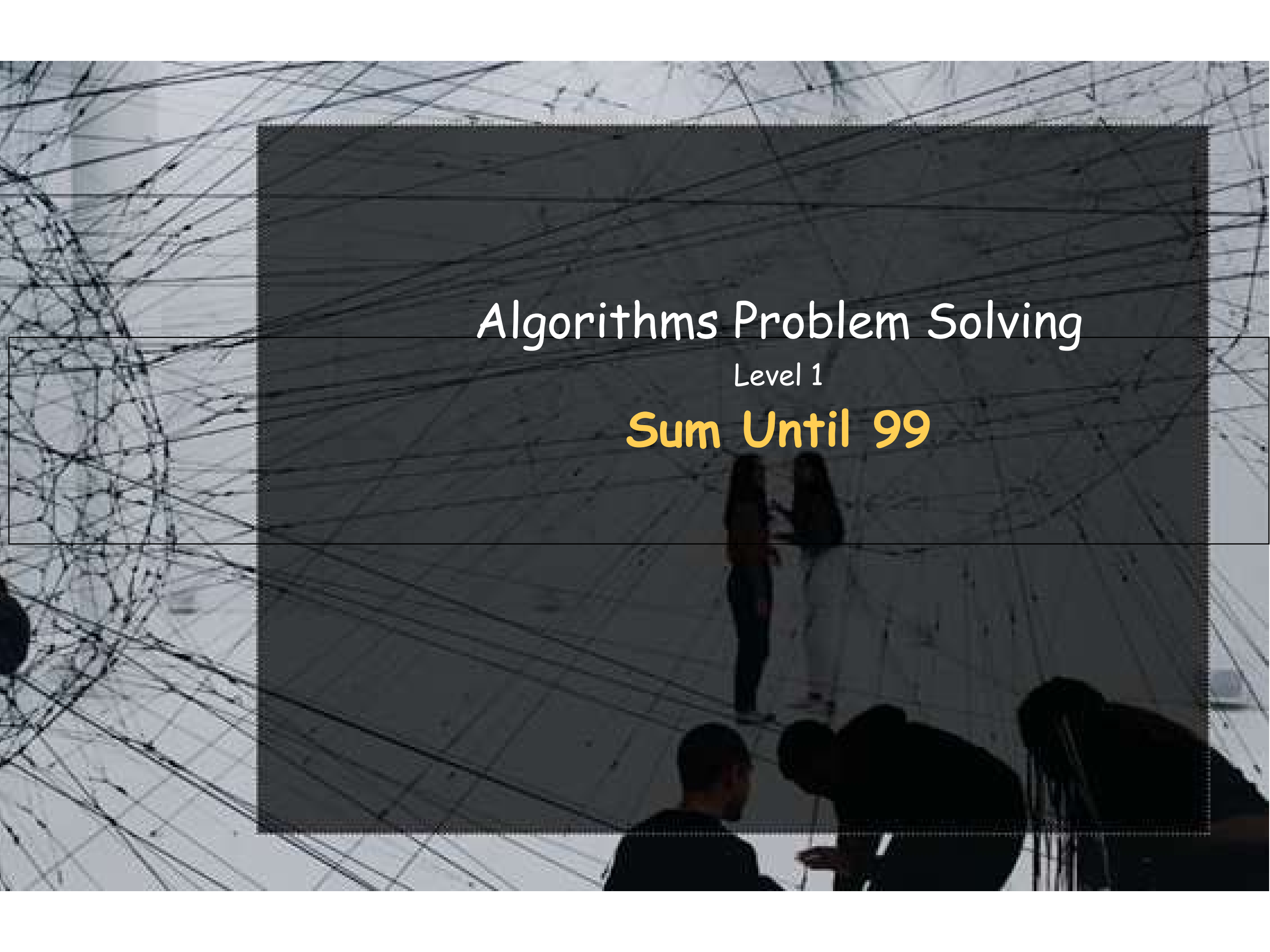
Level 1

Sum Until 99

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Definition

- ◆ This program reads numbers from the user, adds them together, and stops reading when the user enters 99.
- ◆ The number 99 is a sentinel value, which means it is used to stop the loop and is not added to the sum.
- ◆ The program uses functions, a while loop, an if condition, and the break statement.

Explanation

1. Initialize sum to 0.
2. Read a number from the user.
3. Check if the number is 99.
4. If it is 99, stop the loop using break.
5. Otherwise, add the number to sum.
6. Repeat until 99 is entered.
7. Return the final sum.
8. Print the result.

Steps in main()

1. Read the number N.
2. Set `sum` to 0.
3. Start the counter at 1.
4. Check whether the current number is odd.
5. If it is odd, add it to `sum`.
6. Increase the counter by 1.
7. Repeat until the counter reaches N.
8. Return the final sum.
9. Print the result.

💡 Example

◆ If the user enters:

10

20

5

99

◆ The calculation is:

$$10 + 20 + 5 = 35$$

➡ Result

Result = 35

Note

- ◆ 99 is only a stopping value. It is not included in the sum.
- ◆ A sentinel value is a special value used to signal that input should stop.

⊘ Warning

◆ The program uses double for input and calculation, but `printResult()` receives an int. If the sum contains decimals, the decimal part will be lost when passed to `printResult()`.



Comparison

▨ Number is not 99

- ✓ The number is added to sum.
- ✓ The loop continues.
- ✓ Another number is requested.

▤ Number is 99

- ✓ The number is not added to sum.
- ✓ break stops the loop.
- ✓ The final sum is returned.

✦✦ Summary

- ✓ 99 is used as a sentinel value to stop input.
- ✓ break immediately exits the while loop.
- ✓ The sentinel value is not included in the calculation.
- ✓ Functions can separate input, calculation, and output responsibilities.

Thank you 😊